SELECTED RESEARCH CASE STUDIES

Human Factors & UX Research Portfolio

Experimental research, human-technology interaction, and AI-supported experiences

Understand users →	Design the study →	Measure behavior →	Translate evidence

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What this portfolio demonstrates

I use experimental design, mixed methods, behavioral measurement, statistical analysis, and human-factors principles to understand how people interact with complex technologies,and to translate evidence into better experiences, decisions, and systems.

Selected work spans AI-supported learning, human-autonomy teaming, and simulated spaceflight.

Research approach

Across these projects, my work follows a consistent evidence-to-action cycle: define the human problem, translate it into testable questions, collect meaningful behavioral or self-report data, analyze the evidence, and use the findings to inform design or future research.

Human problem & context →	Research questions & measures →	Study execution & data →	Analysis & interpretation →	Design / research implications

3	62	100 / 87
featured case studies	AI-workshop participants	usable dissertation samples

CASE STUDY 01

Evaluating an AI-Supported Learning Experience
StudyHelper / AI-integrated research methods workshops

CASE STUDY 02

Modeling Trust in Human-Autonomy Teams
Soldier-robot teaming / human-machine systems

CASE STUDY 03

Stress & Communication Delay in Simulated Mars Missions
Doctoral dissertation / experimental human factors

CORE METHODS

Experimental design • usability-oriented evaluation • surveys • interviews / qualitative inputs • repeated-measures analysis • regression • physiological measurement • human-autonomy research

Portfolio note: The AI study is presented as a research project/manuscript in preparation rather than as a published paper. The soldier-robot trust study was published at the 2024 IEEE International Conference on Human-Machine Systems; the Mars study is the author's 2021 doctoral dissertation.

CASE STUDY 01

Evaluating an AI-Supported Learning Experience

StudyHelper: using AI chatbots to support undergraduate research-design learning

62	15-30 min	3
students across 3 course sections	chatbot interaction	instructional contexts

THE CHALLENGE

Research methods students often struggle with abstract study-design decisions and confidence in their ability to design and analyze experiments. The project explored whether a brief interaction with purpose-built AI chatbots could support perceived competence.

The study specifically targeted perceived competence rather than assuming that an AI interaction would automatically produce objective learning gains.

THE EXPERIENCE

StudyHelper was designed as an instructor-controlled environment for creating and deploying custom chatbots. The system supported iterative prompt development, proactive conversations, curated response patterns, and sanitized transcripts for instructors.

For the research-methods sections, one chatbot assessed research-design knowledge and another guided students through hypotheses, variables, and study-design choices.

Research question

Does a brief AI-integrated workshop change students’ perceived competence in designing experimental research studies, and does that change correspond with objective knowledge performance?

Role & contribution

The work combined research design, AI-supported learning experience development, instructor co-design, participant data collection, quantitative analysis, and interpretation of both perceived competence and objective knowledge. The workshop was co-designed around course goals and common student difficulties.

01 | Study design & evidence

The evaluation used a repeated pre/post structure across three course sections: two sections of Experimental Research Methods and Inferential Statistics and one Aviation Research section. Students completed a pretest, interacted with the chatbot(s), then completed a posttest and submitted their interaction transcripts and reflections. A subset also participated in a focus group.

Consent + pretest →	15-30 min AI interaction →	Posttest + experience items →	Transcripts + reflections →	Optional focus group

MEASURES

- Perceived competence: single 7-point self-report item
- Six multiple-choice knowledge items
- Additional experience questions
- Chatbot transcripts and written reflections
- Optional focus-group feedback

ANALYSIS

Pre/post perceived-competence and knowledge scores were analyzed with paired-samples t-tests separately by course section. Regression analyses examined whether knowledge performance related to perceived competence.

Perceived competence: pre vs. post

Course section	Pre	Post
Exp. Methods 1	3.38	2.24
Exp. Methods 2	3.31	1.88
Aviation Research	3.39	2.50

Lower scores indicate greater perceived competence on the 1-7 scale.

Perceived competence improved significantly across all three course sections, with large pre-to-post effects (Cohen's d = 0.87–1.19). Objective knowledge increased significantly in only the first Experimental Methods section; the other two sections showed minimal change.

01 | What the evidence means

Key finding

The AI interaction was consistently associated with higher perceived competence, but objective knowledge gains were not consistent across contexts. Perceived competence and knowledge were not significantly related in the reported analyses.

WHAT WORKED

• Brief interaction was feasible within a classroom workshop.
• Perceived competence increased across all three course sections.
• The experience could be adapted across instructors and disciplines.
• Transcript capture created an additional window into learner interactions.

WHAT DID NOT FULLY WORK

• Objective knowledge gains were not uniform.
• Perceived competence was measured with a single self-report item.
• The reported manuscript did not yet analyze the collected chatbot transcripts, reflections, or focus-group data.

Design implications

The findings suggest that confidence and demonstrated knowledge should be treated as related but distinct outcomes when evaluating AI-supported learning experiences. A next iteration should examine the conversational data and use more granular, multi-item measures of competence to identify which parts of the interaction drive perceived improvement.

Why this matters for UX / Human Factors

This project demonstrates an evidence-driven approach to evaluating an AI interaction: define the user problem, co-design the experience with domain experts, observe users interacting with the system, measure outcomes, and use mismatches between subjective and objective outcomes to guide the next design iteration.

Status: Research study / manuscript in preparation

CASE STUDY 02

Modeling Trust in Human-Autonomy Teams

Soldier-robot teaming: connecting psychological trust constructs to human-factors engineering and future system design

2024	3	HITL
IEEE ICHMS publication	trust-model classes	future experimental focus

THE PROBLEM

In soldier-robot teaming, both under-trust and over-trust can create performance and safety problems. The work examined how trust can be modeled and how research can identify the factors needed to support calibrated trust and future trust-management systems.

The paper distinguishes three complementary model types:

Psychological	Design	Mathematical

models →	frameworks →	models

PSYCHOLOGICAL

Used to understand phenomena such as trustingness, perceived trustworthiness, dispositional factors, situational factors, and calibrated trust.

ENGINEERING TRANSLATION

Design frameworks connect human factors to system requirements, tools, guidelines, templates, and outputs that can feed engineering workflows.

My contribution was within the C3 Human Factors team represented on the published work, translating the human-factors research problem into a structured trust-modeling and test-and-evaluation framework.

02 | From human construct to system requirement

A central challenge in this work was avoiding the assumption that a psychological model automatically becomes an engineering model. The paper explicitly treats conceptual models, design frameworks, and mathematical models as complementary rather than interchangeable.

Calibrated trust

Calibrated trust describes the desired alignment between the human's trust and the autonomy's trustworthiness. When system reliability or performance changes, human trust should adjust accordingly.

Human factors & context	Trustingness	Perceived trustworthiness	Calibrated trust	System behavior / adaptation

RESEARCH METRICS

The paper identifies measures such as control takeovers, desire to activate/deactivate the robotic asset, workload questionnaires, physiological workload indicators, and complacency-related measures as potential experimental trust indicators.

DESIGN METRICS

For a future trust-management system, metrics must also represent system characteristics and areas of concern such as safety, dependability, system integrity, and situation awareness.

This is the kind of translation that is central to Human Factors Engineering: psychological evidence becomes useful when it can inform what a system should measure, display, control, or adapt.

02 | Why this is a strong HF case study

HUMAN FACTORS

- Trust in automation
- Human-machine teaming
- Workload and stress
- Situation awareness
- Human error / misuse / disuse
- Human-in-the-loop experimentation

ENGINEERING TRANSLATION

- Requirements thinking
- Trust-management-system concepts
- Design metrics
- Test & evaluation planning
- Interface implications
- Future algorithm inputs / outputs

Portfolio takeaway

I can work at the boundary between behavioral science and engineering,understanding a human construct, decomposing it into measurable factors, and considering how those factors should influence system design and evaluation.

Publication

Fang, Hou, Pavlovic, Cameron, Shirshekar, & Banbury (2024). Trust factors identifying and weighting for trust modeling in soldier-robot teaming. 2024 IEEE 4th International Conference on Human-Machine Systems (ICHMS). DOI: 10.1109/ICHMS59971.2024.10555826.

The paper reports that limited work had defined practical real-time trust metrics for soldier-robot teaming and identifies future human-in-the-loop experimental work as an important path forward.

CASE STUDY 03

Stress & Communication Delay in Simulated Mars Missions

Doctoral dissertation research | Peer-reviewed publication

Experimental human-factors research in a high-stakes, time-sensitive simulated environment

N=100	N=87	45 min
DS inferential sample	physiological SI sample	each mission

THE HUMAN-FACTORS QUESTION

How does a communication delay between crew and mission control affect stress during a simulated Mars mission, and do individual personality differences help explain variation in the stress response?

The study compared a 1-3 second control communication condition with a 2-minute one-way delay. Participants completed critical mission scenarios in a Re-entry space simulator while communicating with mission control.

Simulation orientation →	Control mission →	Break + personality →	Delay mission →	Stress + performance data

SUBJECTIVE MEASURE

A stress Visual Analog Scale was administered before and immediately after each mission. The pre/post difference produced a Difference in Stress (DS) score.

OBJECTIVE MEASURE

Physiological stress was collected throughout each mission with a Polar H10 heart-rate monitor and represented using Baevsky's Stress Index (SI).

03 | Experimental design & analysis

The study used a within-subjects, repeated-measures design. Participants completed two 45-minute, time-sensitive missions, with a 25-minute break between missions for personality and locus-of-control measures. Emergency scenarios and communication transmissions were standardized within the simulation.

PREDICTORS / CONDITIONS

- Communication condition: control vs. 2-min delay
- Big Five personality traits
- Locus of control and demographics as control variables
- Mission / emergency order

OUTCOMES

- Difference in Stress (DS)
- Baevsky Stress Index (SI)
- Post-mission stress VAS
- Perceived communication quality

ANALYTIC STRATEGY

Data screening & assumptions →	Repeated-measures ANOVA →	Supplemental paired t-tests →	Simultaneous regression →	Interpretation & implications

A MANOVA was initially considered, but DS and SI were essentially uncorrelated in both conditions. Separate repeated-measures ANOVAs were therefore used for the two stress outcomes. Simultaneous regression was used to examine personality predictors of stress in the delay condition.

This case demonstrates not only statistical execution but methodological judgment: the analysis plan was adjusted based on the observed properties of the data rather than forcing the originally intended model.

03 | Findings & interpretation

Primary finding

The 2-minute communication delay did not produce a statistically significant increase in either Difference in Stress or physiological Stress Index compared with the control condition.

STRESS OUTCOMES

DS: $F(1,99) < .01$, $p > .99$
SI: $F(1,86) = .02$, $p = .88$

Post-mission stress VAS was also not significantly different between conditions: $t(99) = 1.44$, $p = .15$.

COMMUNICATION QUALITY

Participants did perceive a meaningful communication difference. Communication-quality ratings were significantly lower in the delay condition: $M = 6.05$ vs. 8.48; $t(99) = 9.61$, $p < .001$.

Personality findings

A model containing extraversion and conscientiousness significantly predicted DS in the delay condition, $R^2 = .07$, $F(2,97) = 3.73$, $p = .03$. Extraversion was negatively related to DS while conscientiousness was positively related to DS in that model. The SI model containing agreeableness, conscientiousness, and openness was not significant.

Why the null result matters

The study illustrates an important human-factors principle: a manipulation can demonstrably change the user's experience without producing the hypothesized physiological or stress outcome. Here, participants clearly perceived degraded communication quality, but the measured stress response did not differ significantly. That distinction matters when selecting measures and interpreting system impacts.

Publication

Shirshekar, S., Wheeler, B., & Li, T. (2022). Exploring Personality and Stress during Communication Delays in Simulated Spaceflight Missions. Collegiate Aviation Review International, 40(2), 33–48.

What I bring to research teams

These projects represent different domains, but the underlying capability is consistent: I can move from an ambiguous human problem to a defensible research design, execute studies with real participants, analyze quantitative and qualitative evidence, and communicate what the evidence means for the next decision.

RESEARCH DESIGN

Experimental • repeated-measures • pre/post • mixed methods • surveys • interviews / focus groups • human-subjects research • usability-oriented evaluation

HUMAN FACTORS

Human-automation interaction • trust in autonomy • workload / stress • situation awareness • user behavior • system usability • human-centered design

ANALYSIS
Paired t-tests • ANOVA • regression • descriptive statistics • assumption checking • behavioral and physiological measures

TRANSLATION
Research synthesis • design implications • stakeholder communication • evidence-based recommendations • interdisciplinary collaboration

A note on the portfolio

The case studies intentionally focus on the research problem, decisions, evidence, and implications rather than reproducing academic papers. Detailed citations and full methods are available in the source publications and dissertation.

Additional source materials

• Shirshekar, S. (2021). Exploring the Effects of Communication Delays and Personality on Stress during Simulated Space Missions. Florida Institute of Technology dissertation.

• Shirshekar et al. (manuscript in preparation). Supporting Students' Perceived Competence in Experimental Research Design Through AI Chatbots.